MIL-I-55536A
15 October 1969
Superseding
MIL-I-55536(EL)
26 November 1968

MILITARY SPECIFICATION

IMAGE INTENSIFIER ASSEMBLY 25 MILLIMETER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. - This specification covers the 25 millimeter image intensifier assembly. It is a three stage cascade, fiber optics coupled image intensifier which amplifies low level light from objects or backgrounds.

1.2 Classification.

1.2.1 Types. - This specification covers image intensifier assemblies of the following types:

MX-7854/UV	- Assemblies which contain a window with no reticle over the output screen
MX-7855/UV	- Assemblies which contain no window over the output screen
MX-8501/UV	- Assemblies which contain a window with M-14 reticle over the output screen

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 5855

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SPECIFICATIONS

Federal

PPP-B-566	- Box, Folding, Paperboard
PPP-B-636	- Box, Fiberboard
PPP-B-676	- Box, Set-Up
PPP-C-850	- Cushioning Material, Polystyrene Expanded, Resilent (For Packaging Uses)
PPP-T-60	- Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	- Preservation, Methods of
MIL-F-22191	- Films, Transparent, Flexible, Heat Sealable,for Packaging Applications
MIL-B-43291	- Box, Fiberboard, Corrugated, Double-Wall, Weather Resistant

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U. S. Military Property
MIL-STD-781	- Reliability Tests: Exponential Distribution

DRAWINGS

ECOM

SC-D-611726	- Container, Packaging, Image Intensifier
SC-DL-611731	- Image Intensifier Assembly,(25 MM) MX-8501/UV
SC-DL-611752	- Image Intensifier Assembly, (25 MM) MX-7854/UV

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SC-DL-611755 - Image Intensifier Assembly, (25 MM) MX-7855/UV

SC-DL-611775 - Test Housing Assembly (25 MM)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The image intensifier shall consist of three stages cascaded and assembled in accordance with SC-DL-611731, SC-DL-611752 or SC-DL-611755, as specified. It shall have an S-20 spectral response input photocathode modified with extended red response, an aluminized P-20 type phosphor output image screen, and a high voltage multiplier module requiring 2700 ± 100 volts (peak to peak) at 1200 to 2000 Hz input to develop 45,000 volts direct current (dc) as the nominal operating potential. The image intensifier shall hereinafter be referred to as "intensifier"

3.2 Drawings. - The drawings forming a part of this specification are engineering design drawings. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly,and operation of the intensifier. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.3 Preproduction. - The contractor shall furnish six (6) each intensifiers to prove, prior to starting production, that his production methods and choice of design details will produce intensifiers that comply with the requirements of this specification. The contracting officer shall be notified ten (10) working days prior to submission of the preproduction models for tests and examination. Examination and tests shall be those specified herein (see 4.3) and shall be witnessed by an authorized representative of the contracting officer. Any changes or deviations from the preproduction models during production shall be subject to the approval of the contracting officer. Approval of the preproduction models by the contracting officer shall not relieve the contractor of his obligation to furnish intensifiers conforming to this specification.

3.4 Initial production. - The contractor shall furnish to the Government six (6) intensifiers for inspection as specified in 4.4.

3.5 Material. - Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.5.1 Photocathode. - The photoemissive material of the input photocathode shall have an S-20 spectral response modified with extended red response. The input photocathode fiber optic faceplates of all types of intensifiers shall hereinafter be referred to as the 'photocathode'.

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3.5.2 Image screen. - The sensitive material of the output image forming screen shall be type P-20 aluminized phosphor.

3.6 First stage photocathode sensitivity.- First stage photocathode sensitivity shall be not less than the following values for the condition specified:

a. Luminous sensitivity of 175 microamperes per lumen for 2870° K radiation.

b. Radiant sensitivity of 0.010 ampere per watt at 0.80 micron.

c. Radiant sensitivity of 0.003 ampere per watt at 0.85 micron.

3.7 Vibration. - The intensifier shall not be damaged (see 6.3.1) and shall meet all operational requirements after being rigidly mounted, singly, or in groups, and vibrated with simple harmonic motion parallel to the optical axis (see 6.3.2) of the intensifier and perpendicular to the optical axis of the intensifier over a frequency range of 10 Hertz (Hz) to 55 Hz at an amplitude of not less than 0.10 inch total excursion for 10 minutes in each plane with no voltage applied during the vibration.

3.7.1 Shock. - The intensifier shall not be damaged and shall exhibit no flashing, flickering, or electrical breakdown when subjected to six shock impacts applied perpendicular to the optical axis of the intensifier and six shock impacts applied parallel to the optical axis of the intensifier. The shock impacts shall have a minimum of 75 g's (see 6.3.3) at the peak amplitude.

3.8 Environmental temperature. - The intensifier shall not be damaged by storage, operation, or thermal shock and shall meet specified operational requirements under the following environmental temperature conditions (see figure 1):

a. Storage: - At plus 68° C for 2 hours and minus 54° C for 2 hours

b. Operation: - (1) When subjected to plus 52° C for 30 minutes, and at the end of the 30 minute period with the intensifier operating at plus 52° C and no radiation incident on the photocathode, the intensifier shall have an input capacitance of not greater than 55 pico farads and the steady state charging current shall be not greater than 200 microamperes.

(2) When subjected to minus 54° C for 2 hours and while operating at minus 54° C the intensifier shall have a relative low temperature gain of not less than 24,500 with input radiation of 1×10^{-5} to 3×10^{-5} footcandle.

c. Thermal Shock: - When subjected to thermal shock from plus 52° C to room temperature (see 6.3.4) and from minus 54° C to room temperature.

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3.9 Equivalent background input. - The equivalent background input of the intensifier with no radiation incident on the photocathode shall not exceed 2×10^{-11} lumen square centimeter at room temperature.

3.10 Luminance gain. - The intensifier shall have a luminance gain of not less than 35,000 with input radiation of 1×10^{-5} to 3×10^{-5} footcandles at room temperature.

3.10.1 Maximum output luminance. - The maximum intensifier output luminance shall not exceed 100 footlamberts. This maximum shall occur with input illumination of not less than 1×10^{-3} footcandle.

3.11 Cathode and screen quality. - When the screen of the intensifier is viewed with a 6 to 10 power magnifier with no radiation incident on the photocathode, there shall be no ION spots (see 6.3.5) or field emission (see 6.3.6). When the screen is viewed with a 6 to 10 power magnifier and with the radiation level on the photocathode adjusted to obtain best spot contrast, the opaque or dark spots and bright spots which exceed a contrast of 30 percent of their surrounding area shall not exceed the size and quantities specified in Table I. Size of the non-circular spots shall be determined on the basis of equal area to circular spots. When the distance between two spots is less than the maximum dimension of either spot, the two spots shall be considered as one spot with a size equal to the sum of the maximum dimensions of the two spots plus the amount of separation between them. With the photocathode illuminated, there shall be no bright spots greater than 0.003 inch diameter within the 0.30 inch diameter quality circle and no bright spots visible on the image screen greater than 0.006 inch diameter.

TABLE I CATHODE AND SCREEN SPOTS

SIZE OF SPOTS	Number of Spots Within 0.30 Inch Diameter Circle	Number of Spots Within Area Bounded by Two Circles 0.30 and 0.80 Inch in Diameter	Number of Spots Within Area Bounded by Two Circles 0.80 and 0.984 Inch in Diameter
Greater than 0.015	0	0	0
0.012 to and including 0.015	0	1	2
0.009 to less than 0.012	0	3	8
0.006 to less than 0.009	0	12	26
0.003 to less than 0.006	3	Minimum	Minimum
Less than 0.003	Minimum	Minimum	Minimum

NOTE: The 0.30, 0.80 and 0.984 inch circles on the image screen shall be concentric with the optical axis of the intensifier.

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3.12 Image alinement. - The center of an image produced on the screen of the intensifier by focusing a test reticle on the photocathode concentric with the optical axis shall fall within a 0.060 inch diameter circle on the image screen concentric with the optical axis (see figure 3).

3.12.1 Image shift. - The center of an image produced on the screen of the intensifier by focusing a test reticle on the photocathode concentric with the optical axis shall not shift more than 0.005 inch during 30 seconds of operation.

3.13 Center resolution. - The center resolution of the intensifier shall be not less than 25 line pairs per millimeter (mm) (see figure 4).

3.13.1 Peripheral resolution.- The peripheral resolution of the intensifier shall be not less than 23 line pairs per mm at the four points 7 mm from center of the photocathode (see figure 4).

3.14 Modulation transfer function. - The modulation transfer function (MTF) at the center of the image screen shall be not less than:

a. 90 percent MTF at 2.5 line pairs per mm.

b. 55 percent MTF at 7.5 line pairs per mm.

c. 10 percent MTF at 16 line pairs per mm.

3.15 Center magnification. - The center magnification shall be not less than 0.82 or greater than 1.0 at 1 mm radius from the optical center of the intensifier.

3.15.1 Distortion.- The intensifier distortion from center to edge shall be not greater than 25 percent.

3.16 Useful cathode diameter.- The useful cathode diameter shall be not less than 23 mm.

3.17 Cathode and screen uniformity.- The intensifier image screen brightness uniformity shall not vary more than 4 to 1 over a circular area 20.0 mm in diameter centered on the optical axis of the image screen and variations in screen brightness shall not fall outside the tolerance band shown in figure 2 when the photocathode is uniformly illuminated with 2870° K tungsten lamp radiation. When the screen is viewed with a 6 to 10 power magnifier the background shading shall be uniformly graded with no distinct line or demarcation between the light and dark areas. There shall be no evidence of large mottled areas that appear as multi-fiber shading within the 0.80 inch diameter quality circle and there shall be no evidence of heavy, dark boundary areas between the fiber optic multi-fiber bundles.

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3.18 Breakdown.- With no radiation incident on the photocathode of the intensifier, there shall be no evidence of arcing, flashing, flickering, corona, (see 6.3.7) or any other intermittent or continuous failure when the intensifier is energized with an input voltage of 3000 volts peak to peak for 10 seconds at a barometric pressure of 20.2 inches of mercury and a temperature of minus 54° C

3.19 Mean time between failure.- The intensifier shall have a mean time between failure (MTBF) of 2000 hours.

3.20 Dimensions. - The intensifier shall be manufactured in accordance with the appropriate drawings. Industrial tolerances shall be used except as specified herein (see 3.2).

3.21 Identification marking. - Each intensifier shall be marked for identification in accordance with MIL-STD-130. The marking shall include a coded acceptance date. The first two numbers of the code shall be the last two digits of the number of the year. The last two numbers of the code shall be two digits indicating the calendar week of the year (01 through 52). Reading from left to right (or top to bottom) the code number shall designate the year and week of acceptance in that order.

3.22 Workmanship.- All parts and components of the intensifier shall be free of dirt, grease, oil or other extraneous material and from defects that could impair the performance of the intensifier.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Components and material inspection. - The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (4.3).

b. Initial production inspection(4.4).

c. Quality conformance inspection (4.5).

d. Inspection comparison (4.7).

e. Inspection of preparation for delivery (4.8).

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(82) and record brightness reading. Disconnect for operating potential in

4.3 Preproduction inspection.

4.3.1 Examination. - Each preproduction intensifier shall be examined in accordance with Table II. Presence of one or more defects shall be cause for rejection of that intensifier and may be cause for termination of the examination.

TABLE II EXAMINATION

DEFECT	REQUIREMENT PARAGRAPH
101. Dimensions not as specified	3.20
102. Components missing or not as specified	3.1
103. Identification or special marking missing or illegible	3.21
104. Workmanship not as specified	3.22

4.3.2 Tests. - Following successful completion of the examination specified in 4.3.1 each preproduction intensifier shall be subjected to all the tests in Table III. Failure of any test shall be cause for rejection of that intensifier and may be cause for termination of preproduction testing.

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TABLE III TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
1	2	3
First stage photocathode sensitivity	3.6	4.6.2
Vibration	3.7	4.6.3
Shock	3.7.1	4.6.4
Equivalent background input	3.9	4.6.6
Luminance gain (room temperature)	3.10	4.6.7
Cathode and screen quality	3.11	4.6.8
Center resolution	3.13	4.6.10
Peripheral resolution	3.13.1	4.6.10.1
Modulation transfer function	3.14	4.6.11
Environmental temperature	3.8	4.6.5
Maximum output luminance	3.10.1	4.6.7.1
Image alinement	3.12	4.6.9
Image shift	3.12.1	4.6.9.1
Breakdown	3.18	4.6.15
Cathode and screen uniformity	3.17	4.6.14
Center magnification	3.15	4.6.12
Distortion	3.15.1	4.6.12.1
Useful cathode diameter	3.16	4.6.13
Mean time between failure	3.19	4.6.16

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4.4 Initial production inspection. - When specified (see 3.4) six initial production intensifiers shall be selected at random by the Government from the first production lot being produced from production tooling, shall be examined as specified in Table II, and shall be tested as specified in Table III to determine conformance to the requirements of this specification. The inspection will be performed by the contractor under supervision of a representative of the contracting officer. Acceptance of the initial production intensifiers shall not exclude the remaining intensifiers from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. - Failure of an initial production intensifier to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production intensifier and shall be cause for refusal by the Government to continue acceptance of production intensifiers until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on intensifiers previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all intensifiers accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.4.2 Inspection schedule. - Tables II through VI shall be used for preproduction, initial production and quality conformance inspections as specified in 4.3, 4.4, and 4.5. The first six (6) tests listed in Tables III and IV shall be conducted first and in the order listed. The remaining tests may be performed in any order except for "Mean time between failure" which shall be conducted last.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. - Unless otherwise specified in the contract or purchase order, for purposes of inspection, all intensifiers offered for inspection at one time shall be considered a lot as defined in MIL-STD-105.

4.5.2 Sampling. - Samples shall be selected in accordance with MIL-STD-105.

4.5.3 Examination. - The intensifiers shall be examined in accordance with Table II. Presence of one or more defects shall be cause for rejection.

4.5.4 Tests.

4.5.4.1 Group A, individual tests. - Each intensifier shall be subjected to the tests in Table IV. Failure of any test shall be cause for rejection.

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TABLE IV GROUP "A" TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
1	2	3
First stage photocathode sensitivity	3.6	4.6.2
Vibration	3.7	4.6.3
Shock	3.7.1	4.6.4
Equivalent background input	3.9	4.6.6
Luminance gain (room temperature)	3.10	4.6.7
Cathode and screen quality	3.11	4.6.8
Center resolution	3.13	4.6.10
Peripheral resolution	3.13.1	4.6.10.1
Modulation transfer function	3.14	4.6.11

4.5.4.2 Group "B" sampling. - Samples selected in accordance with 4.5.2 and which have passed the test specified in 4.5.4.1 shall be subjected to the tests in Table V. AQL shall be 1.5 percent defective, inspection level III combined.

TABLE V GROUP "B" TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
1	2	3
Environmental temperature	3.8	4.6.5
Maximum output luminance	3.10.1	4.6.7.1
Image alinement	3.12	4.6.9
Image shift	3.12.1	4.6.9.1
Breakdown	3.18	4.6.15
Cathode and screen uniformity	3.17	4.6.14

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4.5.4.3 Group C, special sampling. - This inspection shall consist of the tests specified in Table VI and shall be performed on sample units that have been subjected to and met the requirements of 4.5.4.1 and 4.5.4.2. Sample units shall be selected in accordance with 4.5.2. AQL shall be 6.5 percent defective, inspection level S-4 combined except for "Mean time between failure"(see 4.6.16).

TABLE VI GROUP "C" TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Center magnification	3.15	4.6.12
Distortion	3.15.1	4.6.12.1
Useful cathode diameter	3.16	4.6.13
Mean time between failure	3.19	4.6.16

4.5.4.3.1 Special sampling for inspection. - Sample intensifiers shall be selected at random for each inspection in Table VI each 30 days. The lot size shall be the number of intensifiers accepted in the preceeding 30 days.

4.5.4.3.2 Special sampling lot failure. - In the event of failure of the special sampling lot the contractor shall immediately report in writing each failure occurrance, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and within 5 working days report the results of the investigation. As soon as practicable the contractor shall submit a report on details of the proposed corrective action on (a) the processes and materials, as applicable, and (b) all intensifiers which were manufactured under the same conditions and which may be considered subject to the same failure. Report shall be forwarded to the responsible technical activity designated in the contract or purchase order through the Government Quality Assurance Representative. After corrective action has been taken an additional sample equal in size to the first sample shall be subjected to the special sampling inspection (all inspections or the inspection which the sample failed, at the option of the Government). Final acceptance and shipment will be withheld until the special sampling reinspection results have shown that the corrective action is effective. When it is the Governments decision that the failure does not affect intensifier performance the special sampling lot will be accepted.

4.6 Inspection procedures.

4.6.1 Test conditions.- Tests shall be conducted in accordance with the test procedures specified herein. Unless otherwise specified the following conditions shall apply:

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a. The radiation source used in the tests shall be a tungsten filament lamp operated at a color temperature of 2870°K plus or minus 50° K.

b. The photometer used for screen brightness measurements shall be a Pritchard Model 1970 PR or equal.

c. The photometer used for screen brightness measurements shall be calibrated against a standard source which has a tungsten lamp, opal glass and filters as specified below.

(1) Tungsten filament lamp operated at 2400° K plus or minus 100° K color temperature.

(2) Corning spectral filters, Nos 3 - 71 and 4 - 67 or equal.

(3) Opal glass to produce uniform, diffuse output.

(4) Output brightness to be 0.5 to 0.7 footlamberts uniformly distributed over an output aperture of not less than 25 mm.

d. The amount of radiation from the source incident on the photocathode for each test shall be the amount specified in the test. Tolerances on specified radiation levels shall be plus or minus 10 percent.

e. Operating potential applied to the dc multiplier shall be 2700 volts peak to peak plus or minus 100 volts at a frequency of 1500 Hz plus or minus 30 Hz.

f. Tests shall be performed at room temperature.

g. Meters used for monitoring lamp current and voltage shall have an accuracy of plus or minus 0.25 percent. Meters used for monitoring intensifier input voltage shall have an accuracy of 0.5 percent.

h. Neutral density filters used in test equipment shall have transmission characteristics within 10 percent of the nominal filter transmission from 0.3 micron to 1.0 micron.

i. All tests shall be performed with the intensifier mounted in a grounded test housing conforming to SC-DL-611775.

j. Environmental test chambers shall maintain specified temperatures within plus or minus 2° C.

4.6.2 First stage, photocathode sensitivity. - This test is to be performed on the first stage module of the intensifier prior to attaching the high voltage multiplier after coupling the cascaded assembly. Prior to conducting the sensitivity

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tests a value shall be determined for combined dark current and leakage current for each first stage module. To determine the dark current and leakage current apply plus 300 volts dc to the anode of the first stage module. With no radiation incident on the photocathode measure total first stage current. This current reading in microamperes is the combined dark current and leakage current. With plus 300 volts dc applied to the anode of the first stage module illuminate a 0.750 inch diameter area centered on the photocathode with 0.01 to 0.02 lumen of 2870° K tungsten lamp radiation. Measure total first stage current and subtract dark current and leakage current. This photocurrent in microamperes divided by the actual input lumens is the 2870° K first stage photocathode sensitivity in microamperes per lumen. With the plus 300 volts dc applied as above, insert a 0.80 micron filter between the photocathode and the 2870° K tungsten source. The 0.750 inch diameter area on the photocathode shall be illuminated with 5×10^{-6} to 10^{-4} watt of 0.80 micron radiation. Measure total first stage current and subtract the leakage and dark current determined above. This photocurrent in microamperes divided by the actual input radiation in watts is the first stage cathode radiant sensitivity at 0.80 micron in microamperes per watt. Remove the 0.80 micron filter and insert the 0.85 micron filter between the photocathode and the 2870° K tungsten source. The 0.750 inch diameter area on the photocathode shall be illuminated with 5×10^{-6} to 10^{-4} watt of 0.85 micron radiation. Measure total first stage current and subtract the leakage and dark current. This photocurrent in microamperes divided by the actual input radiation in watts is the first stage cathode radiant sensitivity at 0.85 micron in microamperes per watt.The 0.80 and 0.85 micron filters shall have the following characteristics:

a. Far infrared blocking out to 4 microns.

b. Peak wavelength of 0.8000 plus or minus 0.0025 micron and 0.8500 plus or minus 0.0010 micron.

c. Bandwidth at the 10 percent points of 0.0125 plus or minus 0.0015 micron.

d. Minimum peak transmission of 50 percent.

Failure to meet the sensitivity requirements specified in 3.6 with either 2870° K, 0.80 micron or 0.85 micron radiation shall constitute failure of this test.

4.6.3 Vibration. - The operating potential shall not be applied to the intensifier during vibration testing. Tolerances on specified frequencies shall be plus or minus 2 Hz and tolerance on total excursion shall be plus or minus 0.005 inch. Prior to beginning vibration testing the intensifier shall be visually inspected for physical damage and missing components. No intensifier shall be vibrated which shows evidence of damage or has missing components. Mount the intensifiers rigidly, singly, or in groups, with the photocathode end up. Subject the intensifiers to simple harmonic motion applied in a plane parallel to the optical axis of the

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intensifiers at a varying frequency of 10 to 55 Hz. In one minute, vary the frequency from 10 Hz to 55 Hz and return to 10 Hz. The amplitude of vibration shall be 0.05 inch (0.10 inch total excursion) in each plane. Repeat this frequency sweep 10 times. At the conclusion of the 10 frequency sweeps, apply the simple harmonic motion to the intensifiers in a plane perpendicular to the optical axis of the intensifiers and repeat the above 10 frequency sweeps. At the conclusion of the vibration testing, inspect the intensifiers for damage (see 6.3.1b). Apply the operating potential and observe for damage (see 6.3.1a). Evidence of damage shall constitute failure of this test.

4.6.4 Shock. - This test shall be conducted in a darkened room. Apply the operating potential to the intensifier for a minimum stabilization period of three minutes. At the end of this stabilization period, with the operating potential applied and no light incident on the photocathode subject the intensifier to six (6) shock impacts parallel to the optical axis and six (6) shock impacts perpendicular to the optical axis. Apply the shock impacts in such a way as to generate nominal half-sine wave pulses having a minimum peak amplitude of 75 g's on the intensifier. The duration of each shock pulse shall be 6 milliseconds ± 2 milliseconds measured between the 10 percent values of peak amplitudes. The energy under the shock curve shall be not less than 0.25 g-seconds and the after oscillations shall be not greater than 15 percent of peak amplitude of the nominal half-sine wave pulse. Internal flashing, flickering, or electrical breakdown or evidence of damage during or at the conclusion of the shock test shall constitute failure of this test.

4.6.5 Environmental temperature. - Paragraph 4.6.1a does not apply for environmental temperature testing. Place the intensifier in a test chamber and gradually (in not less than 1/2 hour) raise the temperature of the chamber to plus 68° C and hold at this temperature for a minimum of 2 hours. At the end of the 2 hour stabilizing period gradually (in not less than 15 minutes) lower the chamber temperature to plus 52° C and hold this temperature for 30 minutes. At the end of the 30 minute stabilizing period, with no radiation incident on the photocathode, apply the operating potential to the intensifier. With the operating potential applied,project 100 to 150 footcandles of uniformly distributed radiation on the photocathode for 10 seconds. At the end of the 10 second period of illumination, reduce the light input to 1×10^{-5} to 3×10^{-5} footcandle for a period of 10 minutes and observe the intensifier for damage. After observing for damage exclude all incident radiation from the photocathode. With the operating potential applied and test chamber temperature maintained at plus 52° C the intensifier input capacitance and charging current shall be determined using the grounded housing and test circuit shown in figure 5. (NOTE: All meters, the AC source and other measuring devices are to be mounted outside the test chamber). Care should be taken to keep the stray capacitance of the test housing to ground as low as possible. The capacitance of the input lead to the test housing and return wire should also be accounted for by measuring the voltage across R_1 (see figure 5) when there is no tube in the test housing and subtracting this voltage from the voltage reading taken when the intensifier is in the testing housing. The input capacitance (C) is calculated by determining e_{p-p} and E_{p-p} (see figure 5) and substituting these values in the following equation:

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$$C = \frac{\frac{e_{p-p}}{1000}}{9420 \quad E_{p-p}}$$

The charging current (i_c) is calculated by determining e_c and substituting this value in the equation:

$$i_c = \frac{e_c}{1000}$$

The value of the charging voltage (e_c) is obtained by measuring the peak to peak amplitude, of the spikes present on the e_{p-p} voltage wave-form when viewed on the oscilloscope (see figure 5). The value of e_c shall be determined with the capacitance divider ground connection removed. At the conclusion of the above measurements lower the ambient temperature of the intensifier from plus 52° C to room temperature in not more than 5 minutes. After not less than 1 hour at room temperature operate the intensifier and observe for damage. Place the intensifier in the test chamber with the ambient temperature of the chamber at 23° C. Apply the operating potential to the intensifier and illuminate the photocathode with 1×10^{-5} to 3×10^{-5} footcandle of tungsten lamp radiation. Measure and record the screen brightness (B_1) with a photometer (see 4.6.1b). Lower the temperature of the chamber in not less than 30 minutes to minus 54° C and hold at this temperature for 2 hours. At the end of this 2 hour stabilizing period with no radiation incident on the photocathode apply the operating potential to the intensifier. With the operating potential applied project 1×10^{-5} to 3×10^{-5} footcandle of uniformly distributed radiation on the photocathode, measure the screen brightness (B_2) and record brightness reading. All conditions (input illumination, operating voltage and photometer settings) for brightness measurements, B_2, shall be identical to the conditions prevailing for brightness measurement, B_1. Determine the relative low temperature gain as follows:

$$\frac{B_2}{B_1} \quad X \quad \text{(Room Temperature gain as measured in 4.6.7)}$$

Raise the ambient temperature of the chamber to room temperature in not more than 5 minutes. After not less than 1 hour at room temperature operate the intensifier and observe for damage. Evidence of damage or failure to meet the requirements of 3.8 shall constitute failure of this test.

4.6.6 Equivalent background input. - With the operating potential applied to the intensifier and no radiation incident on the photocathode, hold for a stabilizing period of not more than 30 minutes. At the end of the stabilizing period with no radiation incident on the photocathode measure the screen brightness (footlamberts)

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with a photometer and record photometer readings (R_1). In the event the readings should fluctuate observe the readings for 30 seconds and record the maximum and minimum readings. Illuminate the photocathode with 2 X 10^{-11} lumen per square centimeter uniformly distributed over a 0.750 inch diameter aperture centered on the photocathode faceplate and record photometer reading (R_2). In the event this reading should fluctuate observe the reading for 30 seconds and record maximum and minimum readings. The angle of incident flux shall be not greater than 10 degrees and the acceptance angle of the photometer shall be not greater than 2 degrees. The photometer shall be positioned such that the acceptance angle covers a 14 mm plus or minus 1 mm diameter area centered on the illuminated area of the screen. Average the maximum and minimum dark readings, R_1, and the maximum and minimum illuminated readings, R_2 and determine equivalent background input (E.B.I.) by the following formula:

$$EBI = \frac{R_1}{R_2 - R_1} \quad X \text{ ACTUAL INPUT ILLUMINATION}$$

Equivalent background input greater than that specified in 3.9 shall constitute failure of this test.

4.6.6.1 An alternate method for measuring EBI may be used in lieu of that specified in 4.6.6. When used the following method shall apply:

$$EBI = \frac{I_1 - I_0}{I_2 - I_1} \quad X \quad 2 \times 10^{-11}$$

In this formula:

I_0 = Dark current of photomultiplier tube

I_1 = Photomultiplier current due to brightness of intensifier with no incident radiation

I_2 = Photomultiplier current due to brightness of intensifier when illuminated with 2 X 10^{-11} lumen/cm^2 .

Equivalent background input greater than that specified in 3.9 shall constitute failure of this test.

4.6.6.2 An additional alternate method for measuring EBI which may be used in lieu of that specified in 4.6.6 and 4.6.6.1 is as follows: With the operating potential applied to the intensifier and no radiation incident on the photocathode, hold for a stabilizing period of not more than 30 minutes. At the end of the stabilizing period with no radiation incident on the photocathode measure the screen brightness (footlamberts) with the photometer. Divide this reading by the luminance gain and multiply the result by 1 X 10^{-3}. This figure is the EBI. Equivalent background input greater than that specified in 3.9 shall constitute failure of this test.

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4.6.7 Luminance gain (room temperature).- Illuminate the outside surface of the photocathode with 1 X 10^{-5} to 3 X 10^{-5} footcandle uniformly distributed over a 0.750 inch diameter aperture centered on the photocathode fiber optic faceplate. Apply the operating potential to the intensifier and measure the image screen luminance with the photometer. The acceptance angle of the photometer shall be two (2) degrees or less. Position the photometer so that the acceptance angle subtends a 14 mm plus or minus 1 mm diameter area centered on the illuminated area of the screen. Luminance gain is determined by dividing the screen luminance in footlamberts by the actual input illumination in footcandles. Luminance gain less than that specified in 3.10 shall constitute failure of this test.

4.6.7.1 Maximum output luminance.- Determine the maximum output luminance in a similar manner and with the same equipment as specified in 4.6.7. With the operating potential applied gradually increase the input radiation level from 1 X 10^{-5} footcandle until maximum luminance is reached. Input radiation shall be uniformly distributed over the photocathode faceplate. Maximum output luminance at an input radiation level below that specified in 3.10.1 or screen luminance greater than that specified in 3.10.1 shall constitute failure of this test.

4.6.8 Cathode and screen quality. - With the operating potential applied and no radiation incident on the photocathode, observe the image screen with a 6 to 10 power magnifier. The screen shall be observed for the defects listed in Table VII. Gradually increase the light level and continue to observe the screen for the defects listed in Table VII. With the light level adjusted for best spot contrasts the image screen shall be observed for opaque or dark spots and bright spots which exceed the size or quantity specified in Table I and for defects listed in Table VII. The axis of the viewing screen shall be perpendicular to the output screen within 3 degrees. Failure to meet all the requirements of 3.11 shall constitute failure of this test.

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TABLE VII CATHODE AND SCREEN DEFECTS

DEFECTS	NO CATHODE ILLUMINATION	CATHODE ILLUMINATION	SCREEN LOCATION
Strong ion spot (see 6.3.5)	Diffuse area	Diffuse area	Center
Moderate ion spot (see 6.3.5)	Not visible	Diffuse area	Center
Weak ion spot (see 6.3.5)	Not visible	Cluster of scintillations (see 6.3.8) not to be confused with electron scintillations (see 6.3.9)	Center
Field emission (see 6.3.6)	Bright spots or patterns spatially fixed	Generally Disappear	Anywhere on the screen
Dark boundaries between multi-fiber bundles	Not visible	Geometrically shaped figures caused by heavy boundaries between the multi-fibers in the fiber optic plate	Generally uniformly distributed over entire screen
Corona (see Note 1) (see 6.3.7)	Bright spots generally spatially fixed	Bright spots, generally spatially fixed	Anywhere on the output screen
Leakage or faceplate breakdown (see Note 2)	Appears as heavy field emission over general field with one or several bright spots spatially fixed generally accompanied with a frying or snapping sound	Bright spots appearing either intermittently or continuously and brightening or flashing of general screen background	Anywhere on the output screen

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NOTE 1 TO TABLE VII

To distinguish between field emission and corona, place the intensifier in a housing which is sealed with a window. Repeat 4.6.8. Disappearance of the bright spots indicate the phenomenon was corona.

NOTE 2 TO TABLE VII

Immediately remove operating potential and discontinue testing at the first indication of leakage or breakdown.

4.6.9 Image alinement. - With the operating potential applied focus the test pattern described in figure 3 on the photocathode such that the center dot falls on the optical axis of the photocathode. The photocathode shall be illuminated to provide a high contrast image of the test pattern. Observe the image of the test pattern formed on the screen of the intensifier with a 10 power (or higher) measuring microscope containing a reticle of 0.060 inch apparent diameter. The microscope shall be alined with the optical axis of the intensifier such that the center of the 0.060 inch diameter reticle falls on the optical axis. The center of the test pattern shall fall within the 0.060 inch diameter circle. Failure to meet the requirements of 3.12 shall constitute failure of this test.

4.6.9.1 Image shift. - Determine image shift with the same equipment used in 4.6.9. At the conclusion of the image alinement test (4.6.9), position the center of the reticle in the measuring microscope on the center of the test reticle and remove the operating potential from the intensifier. After the intensifier is completely discharged, apply the operating potential and observe the test reticle for 30 seconds. Should the test reticle be displaced from its original position (prior to removing the operating potential) measure the displacement. Image shift greater than that specified in 3.12.1 shall constitute failure of this test.

4.6.10 Center resolution. - Perform this test using:

a. A radiation source (see 4.6.1a).

b. A projection system having an f number not greater than 10 to project the test pattern such that the size of the pattern on the photocathode of the intensifier is the same as the pattern described in figure 3.

c. A resolving power target (see figure 4),or the required sections of the 1951 Air Force Resolving Power Target.

d. A 6 power or higher viewing system.

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The resolving power target shall be focused or butted on the photocathode such that the center of the target is alined with the optical axis. Input radiation level shall be adjusted for best image contrast. The image of the resolving power target formed on the screen of the intensifier shall be observed for limiting resolution (see 6.3.10). Resolution less than that specified in 3.13 shall constitute failure of this test.

4.6.10.1 Peripheral resolution. - Determine peripheral resolution with the same equipment used in 4.6.10. The resolving power target shall be focused or butted on the photocathode such that the center of the target is positioned 7 mm from the optical center of the photocathode. Input radiation shall be adjusted to provide best image contrast. The image of the resolving power target formed on the screen of the intensifier shall be observed for limiting resolution. Resolution less than that specified in 3.13.1 shall constitute failure of this test.

4.6.11 Modulation transfer function (see 6.3.11). Perform this test using the modulation transfer function analyzer system furnished by the Government. Prior to conducting MTF tests a correction factor will be required for each spatial frequency of interest to take into account system degradation. This correction factor shall be determined for the MTF Analyzer System as follows:

The theoretical spatial frequency response of a 100 micron slit is supplied with each MTF Analyzer System. From this theoretical response, the MTF values at 2.5 lp/mm, 7.5 lp/mm and 16 lp/mm is recorded. Using the calibration slit assembly, an MTF curve is run on the system after it has been adjusted to obtain optimum readings. The system response is then determined for the spatial frequencies of 2.5 lp/mm, 7.5 lp/mm, and 16 lp/mm .

To obtain the correction factor for each of the spatial frequencies take the difference between the theoretical MTF response and the actual system MTF response at the points of 2.5 lp/mm, 7.5 lp/mm and 16 lp/mm.

Place the slit assembly on the photocathode of the intensifier assembly and illuminate the slit with 5×10^{-3} footcandle. Apply the operating potential to the intensifier and aline the output slit so that its center is alined with the optical axis of the MTF Analyzer Head

Adjust the input optics on the analyzer head for the best image as viewed through the eyepiece. Recheck the focus by adjusting the fine focus control to obtain maximum system signal response at the high spatial frequencies.

The MTF at each of the spatial frequencies of interest is determined by the actual system reading plus the correction factor for that frequency.

Intensifier MTF less than that specified in 3.14 at either 2.5 lp/mm, 7.5 lp/mm, or 16.0 lp/mm shall constitute failure of this test.

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4.6.12 Center magnification. - Perform this test with the same equipment used in 4.6.9. With operating potential applied to the intensifier focus the center of the test pattern on the optical axis of the photocathode and adjust input radiation level for best image contrast. The size of the pattern focused on the photocathode is to be the same as the pattern described in figure 3. Measure the chordal separation of the one (1) mm graduations (M_1) as imaged on the screen of the intensifier with a 10 power (or higher) traveling microscope. Compute center magnification as follows:

$$\text{CENTER MAGNIFICATION} = \frac{M_1}{2.00\ (\text{Chordal Separation})}$$

Failure to meet the requirements of 3.15 in both the vertical and horizontal directions shall constitute failure of this test.

4.6.12.1 Distortion.- Perform this test with the same equipment used in 4.6.9 and in the same manner as in 4.6.12. Measure the chordal separation of the 10 mm graduations (M_2) as imaged on the screen of the intensifier. Compute the distortions as follows:

$$\text{PERCENT DISTORTION} = \frac{\left(\frac{M_2}{20.00}\right) - \left(\frac{M_1}{2.00}\right)}{\frac{M_1}{2.00}} \times 100$$

Distortion greater than that specified in 3.15.1 shall constitute failure of this test:

WHERE: a. 20.00 is the separation in millimeters of the test points on the photocathode.

b. M_2 is the measured separation in millimeters of the test points as imaged on the output screen of the intensifier.

Failure to meet the requirements of 3.15.1 in both the vertical and horizontal directions shall constitute failure of this test.

4.6.13 Useful cathode diameter. - Perform this test with the same equipment used in 4.6.9. With the operating potential applied to the intensifier, focus the test pattern on the optical axis of the photocathode and adjust input radiation level for best image contrast. The size of the pattern focused on the photocathode is to be the same as the test pattern described in figure 3. The image screen shall be viewed with a 6 to 10 power microscope. Useful cathode diameter is determined by the number of millimeter graduations visible on the screen of the intensifier and shall be determined in both a vertical and horizontal direction. Failure to meet the requirements of 3.16 in both the vertical and horizontal directions shall constitute failure of this test.

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4.6.14 Cathode and screen uniformity.- Measure the intensifier screen uniformity by uniformly illuminating the photocathode with 1 X 10^{-5} to 3 X 10^{-5} footcandle of 2870° K tungsten lamp radiation and scan the image screen with a spot brightness scanner. The spot brightness scanner shall view the image screen such that a 0.040 inch (or 1 millimeter) diameter area of the screen is observed at any one time. The scanner shall begin its scan at the optical center of the image screen and move in a spiral motion to the edge of the illuminated area of the screen. The spiral scan shall be such that the entire screen area is viewed by the 0.040 inch spot. Correction factors shall be generated for any non-uniformities introduced by either the input illumination system or the scanning devices. The scanner shall travel from center to edge at a rate of 1.0 mm per second or less. Spot brightness shall be measured with an eye-corrected, high sensitivity, fast response detector, and the detector output shall be displayed on an X-Y or strip chart recorder. The spot brightness reading at the center of the intensifier shall be normalized to 4.0. Replace the scanning device with a 6 to 10 power magnifier and view the image screen for the requirements specified in 3.17. Variation in screen brightness greater than 4 to 1 from the center of the screen to points 10.0 millimeters from the center or variations that fall outside the tolerance band shown in figure 2 or evidence of the defects specified in 3.17 shall constitute failure of this test.

4.6.15 Breakdown. - Perform this test at a temperature of minus 54° C and a barometric pressure of 20.2 inches of mercury. Energize the intensifier with a sine wave input potential of 3,000 volts (peak to peak) at 1500 Hz. With no radiation incident on the photocathode, monitor the image intensifier input current and observe for spikes, non-uniform wave shape and other indications of corona or breakdown. Failure to meet the requirements of 3.18 shall constitute failure of this test.

4.6.16 Mean time between failure. - Intensifier MTBF testing shall be performed each 30 days on sample intensifiers selected at random from the previous 30 day production period. The sample size shall be a minimum quantity of 10. For preproduction life testing no replacements are permitted; for production life testing, replacement of failed intensifiers is required. Sample intensifiers selected for life testing shall be selected from lots which have passed the quality conformance inspection of 4.5.3, 4.5.4.1, and 4.5.4.2. Life testing shall be conducted in a darkened area at room temperature with the intensifiers mounted in grounded metal housings. The operating potential shall be applied and there shall be no radiation incident on the photocathode. The intensifiers shall be operated intermittently as follows:

a. On time 55 minutes

b. Off time 5 minutes

c. During the off period, the photocathode will be illuminated with 1 X 10^{-4} to 8 X 10^{-4} footcandle.

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The total number of operating hours for all units, failed and unfailed, original and replacement, shall be monitored and recorded. Once each 24 hours the following inspections shall be performed:

NOTE 1 - During weekends and other periods of general cessation of plant activity, this time period may be extended not to exceed 96 hours.

d. Screen brightness. - After one (1) hour of operation the intensifier(s) image screen brightness shall be measured with a photometer as specified in 4.6.7. There shall be no light incident on the photocathode. This brightness reading shall be recorded as the initial screen brightness. At the end of each 24 hour period the screen brightness shall again be measured. When screen brightness varies more than 30 percent from the initial brightness, the intensifier(s) shall be removed from the life test and a complete equivalent background input test 4.6.7 shall be performed. Failure to meet the requirements of 3.9 shall constitute intensifier failure. When the intensifier meets the requirements of 3.9 it shall be placed back on life test.

e. Cathode and screen quality. - Observe the intensifier with a 6 to 10 power magnifier, with no light incident on the photocathode, for ion spots and field emission. The cathode shall then be uniformly illuminated and the screen observed for ion scintillation and bright spots. Failure to meet the requirements of 3.11 shall constitute failure of this test.

f. Cathode and screen uniformity. - Observe the intensifier with the cathode uniformly illuminated for non-uniformities in screen brightness and mottled areas. Failure to meet the requirements of 3.17 except the luminance scan shall constitute failure of this test.

g. Evidence of breakdown or intermittent operation. - Observe the intensifier for flickering, flashing, blanking, breakdown or any type, snapping, or frying noises or evidence of corona on the output faceplate. Appearance of any of these disturbances shall constitute failure of this test. The test shall be accomplished in accordance with MIL-STD-781, Test Level A1, Test Plan VI. In the event of failure of this test, 4.5.4.3.2 shall apply.

4.7 Inspection comparison. - When initial production inspection is specified, the Government may select intensifiers at any time during the contract production period and subject these intensifiers to the examination and tests specified in 4.4 to determine that the quality of the selected intensifiers is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Intensifiers will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction and initial production intensifiers.

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4.7.1 Inspection failure. - Failure of any intensifier when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of intensifiers until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition that caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. - For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling . - Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. - Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective, inspection level II.

105. Materials not as specified for level A.

106. Each intensifier not preserved, cushioned, and placed in a box as specified for level A.

107. Intensifier not intermediate packaged as specified in level A.

108. Each intensifier not cushioned and placed in a container as specified for level C.

109. Marking illegible, incorrect, or incomplete for levels A, B, and C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A. or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. - Each image intensifier assembly 25 millimeter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each image intensifier assembly 25 millimeter shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein. Each intensifier shall be individually

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packaged method 1A8 as follows: A protective end cap shall be placed over the photocathode and screen of each intensifier, see SC-DL-611731. Cover the high-voltage connector opening with tape conforming to PPP-T-60. Place each item within a bag fabricated of material conforming to MIL-F-22191 type 1, evacuate the air and heat seal. Cushion the bagged item within a two piece interlocking block of expandable polystyrene conforming to PPP-C-850 and in accordance with SC-D--611726. The blocks shall be taped around the seams with tape conforming to PPP-T-60. Place the cushioned item within a close-fitting paperboard carton conforming to PPP-B-566, variety 2 or PPP-B-676. Box closure shall be as specified in the appendix of the applicable box specification.

5.1.1.4.1 Intermediate container. - A quantity of 12 each items packaged as specified in 5.1.1.4 shall be placed within a close-fitting fiberboard box conforming to PPP-B-636, W6c weather resistant. Box closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. - Each image intensifier assembly 25 millimeter shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. - A quantity of intermediate containers packaged as specified in 5.1.1.4.1 shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, V3c or MIL-B-43291 class weather resistant. Box closure shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. - A quantity of intermediate containers packaged as specified in 5.1.1.4.1 shall be packed in accordance with 5.2.1.

5.2.3 Level C . - A quantity of 25 mm image intensifiers shall be packed as specified in 5.2.1, except that the fiberboard boxes shall be the type, size and kind commonly used for the purpose , and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. - The image intensifier covered by this specification is intended for use in an electro-optical viewing device to intensify low level light such that a visible image is presented for viewing and sighting purposes.

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6.2 Ordering data. - Prccurement documents should specify the following:

a. Title, number and date of this specification.

b. Type of intensifier required (see 1.2.1).

c. Level of preservation and packaging and level of packing (see 5.1 and 5.2).

6.3 Definition of terms

6.3.1 Damage. - Damage is defined as:

a. Electrical failure or malfunctioning including arcing,corona, flashing, flickering or blanking.

b. Cracks, breakage, deformation, corrosion, or deterioration of any part or finish, or missing and loose components.

c. Mottled areas, blotches, dark spots, or other visual indications of module separation or degradation of the coupling media.

6.3.2 Optical axis. - Optical axis is defined as the mean center line of the cylinder formed by the outisde diameter of the intensifier housing.

6.3.3 "g". - "g" is defined as the force which results in an acceleration or deceleration of 32.17 feet per second per second.

6.3.4 Room temperature. - Room temperature is defined as 23° C plus or minus 3° C.

6.3.5 Ion spots. - An ion spot appears as a diffuse area or spot of higher brightness in the center of the field. It increases in intensity as the cathode illumination is increased and is usually only visible when the photocathode is illuminated. Its diffuse appearance distinguishes it from other bright defects.

6.3.6 Field emission. - Field emission is an undesirable extraneous emission (excluding thermionic emission) which appears as bright spots or patterns that flicker or appear intermittently on the image screen in one general position. Field emission is voltage dependent and is best observed with no radiation incident on the photocathode.

6.3.7 Corona on output faceplate. - Appear as bright spots in field which are still visible when input exceeds maximum luminance. Occur both intermittently and continuously in position and time. Generally do not occur immediately on turn-on but some short time later as faceplate charges. As conditions become worse, spots glowing other than green appear and can cause breakdown in faceplate.

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6.3.8 Ion scintillations. - Bright spots which are much brighter and larger than electron scintillations. Very easy to resolve and occur in the center of the field, generally in clusters. These bright spots are constantly moving in an intermittent random fashion and are best observed with the photocathode illuminated. One is actually observing the emission of large numbers of electrons from a single ion impact with the photocathode.

6.3.9 Electron scintillations. - Very small bright spots of light which vary in position and time. Usually visible only when well dark adapted and intensifier is not illuminated. When present are far too numerous to count and appear as a scintillating uniform field. These account for the major average background brightness under conditions of no illumination. One is actually observing each separate electron emission event at the photocathode.

6.3.10 Limiting resolution. - Limiting resolution is defined as the smallest resolution pattern which the observer can see and distinguish between the black lines and the clear area between the black lines. The observer must be able to determine the number of line pairs in the test pattern and the direction of the line pairs in both the vertical and horizontal test pattern.

6.3.11 Modulation transfer function (MTF). - A measure of the performance of the image intensifier assembly as an image-producing device. In effect it is the frequency response of the assembly obtained by measuring the degradation of a slit image as it appears on the output screen, as correlated to a variable frequency sine-wave pattern.

Custodian:
Army-EL

Review Activities:
Army-MU, WC

User Activities:
Army-MD

Preparing Activity:
Army-EL

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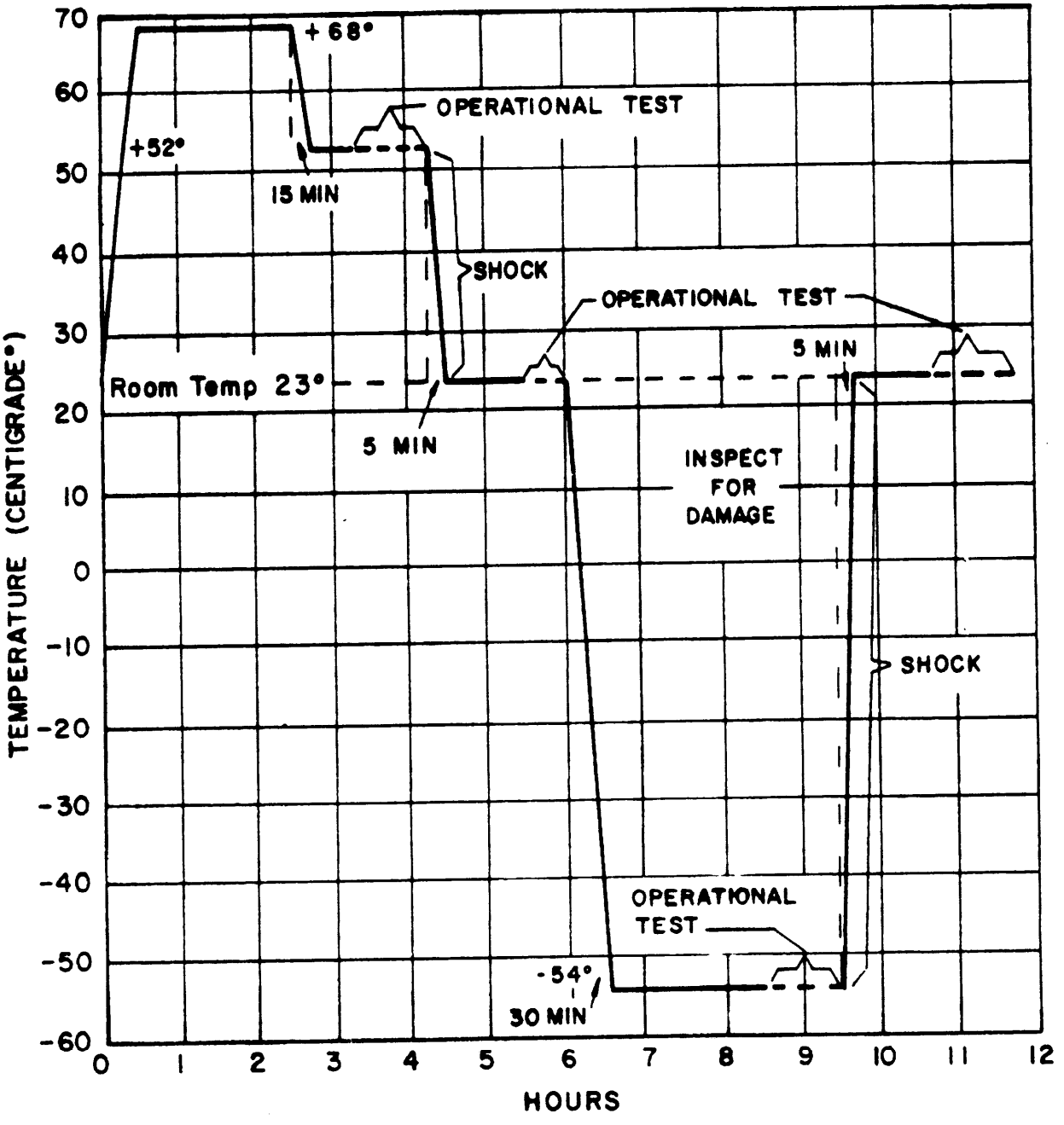
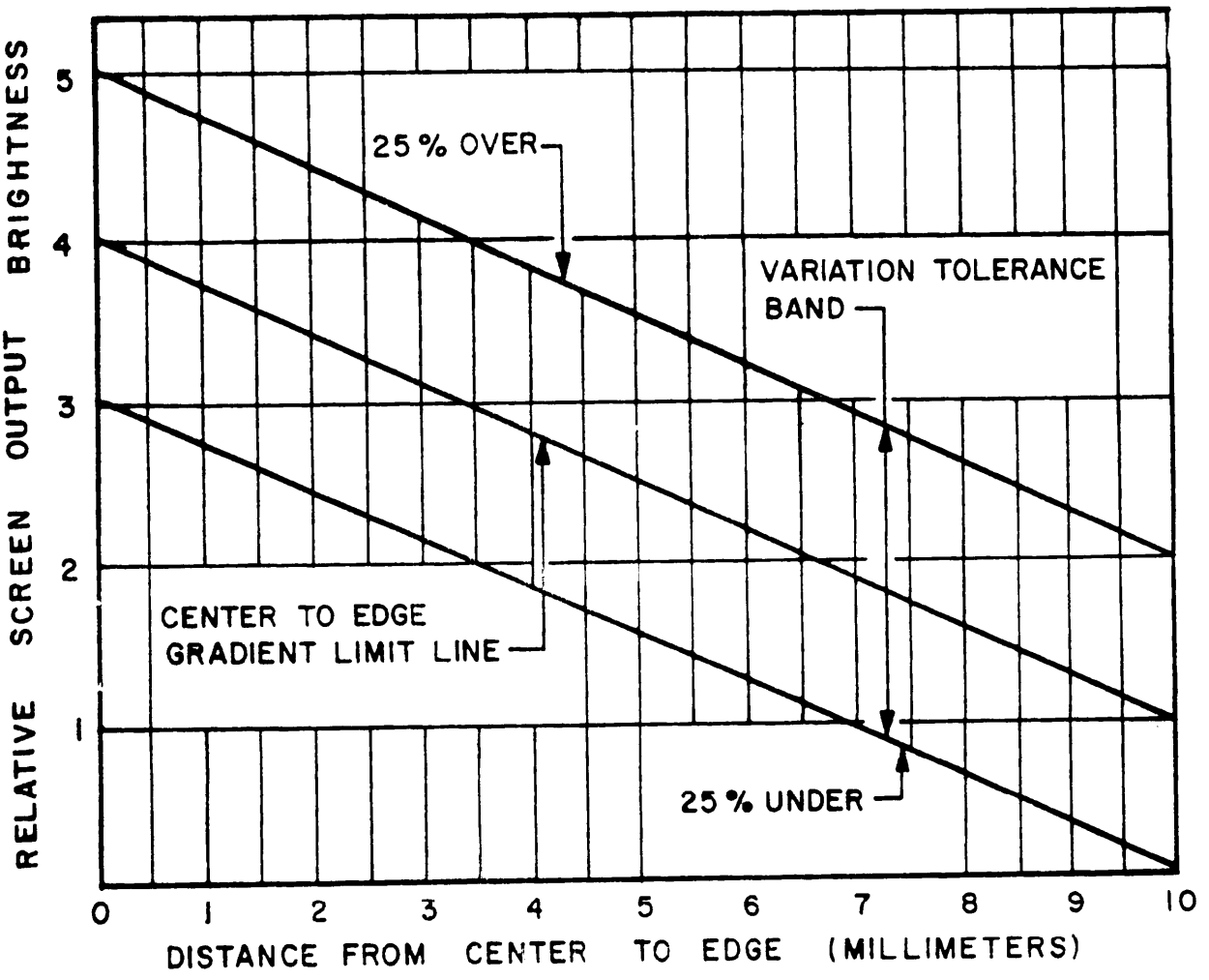


FIGURE 1
ENVIRONMENTAL CYCLE
25mm ASSEMBLY

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CATHODE AND SCREEN UNIFORMITY ACCEPTANCE LIMITS



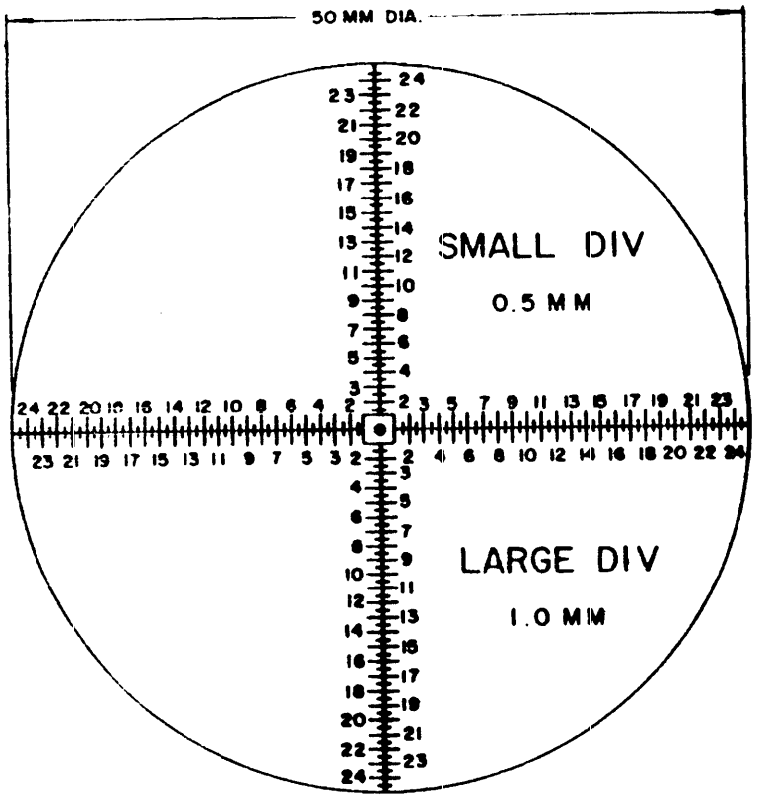
NOTE: THE VARIATION TOLERANCE BAND IS DETERMINED BY DRAWING TWO LINES PARALLEL TO A STRAIGHT LINE GRADIENT FROM 4.0 IN CENTER TO 1.0 AT 10 MM FROM CENTER. ONE LINE IS 25 % ABOVE (OR AT 5.0) AND THE OTHER 25 % UNDER (OR AT 3.0).

FIGURE 2 25mm ASSEMBLY

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NOTES

1. THE TEST RETICLE SHALL CONSIST OF EQUALLY SPACED LINES FROM THE CENTER OF THE 50 MM DIAMETER CIRCLE TO THE EDGE OF THE CIRCLE, IN 4 DIRECTIONS 90° APART. SPACING BETWEEN A LARGE GRADUATION AND A SMALL GRADUATION SHALL BE 0.5 MILLIMETER ± .03 MILLIMETER SPACING BETWEEN TWO LARGE GRADUATIONS SHALL BE 1.0 MILLIMETER ± .03 MILLIMETER. THE CENTER OF THIS TEST RETICLE SHALL BE A SQUARE 2 MILLIMETER ON A SIDE WITH A 0.1270 MILLIMETER DIAMETER SPOT CONCENTRIC WITH THE CENTER OF THE TEST RETICLE. WIDTH OF ALL LINES TO BE 0.127 MILLIMETER ± 0254 MILLIMETER. ALL LINES, LETTERS AND NUMBERS SHALL BE HIGH CONTRAST, BLACK ON A CLEAR GLASS SUBSTRATE 2 INCHES WIDE X 2 INCHES LONG X .060 TO 10 INCHES THICK. THE LETTERING SHOWN IN THE FIRST AND FOURTH QUADRANTS SHALL BE AS LARGE AS POSSIBLE WITHOUT INTERFERING WITH THE GRADUATIONS AND NUMBERS AND SHALL FALL WITHIN A CIRCLE 10 MILLIMETER RADIUS, WHICH IS CONCENTRIC WITH THE 50MM DIA.



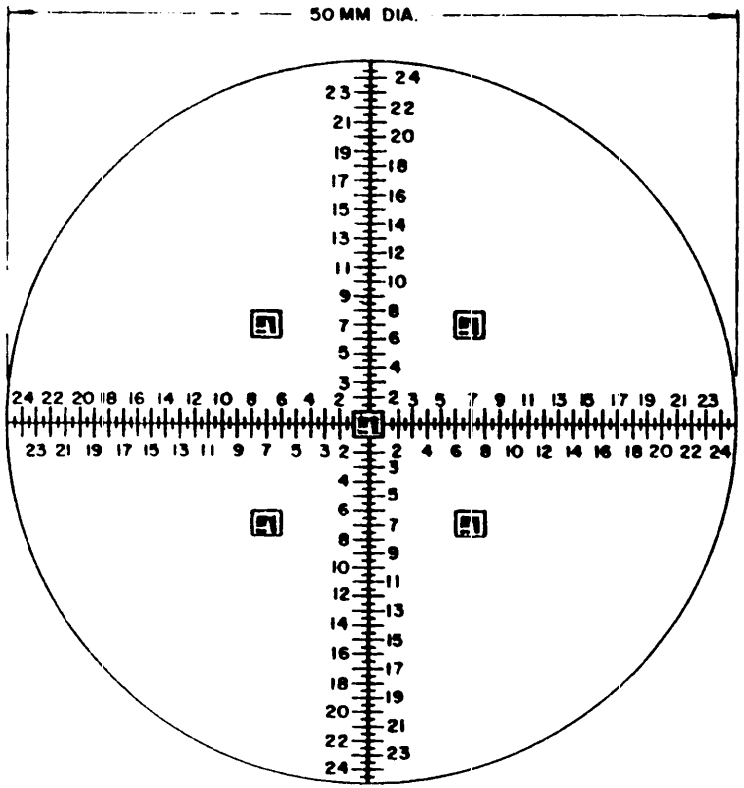
25mm ASSEMBLY

FIG. 3 MAGNIFICATION, IMAGE ALINEMENT AND USEFUL CATHODE DIAMETER TEST RETICLE

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NOTES:

1. THE TEST RETICLE SHALL CONSIST OF EQUALLY SPACED LINES FROM THE CENTER OF THE 50 MM DIAMETER CIRCLE TO THE EDGE OF THE CIRCLE, IN 4 DIRECTIONS 90° APART. SPACING BETWEEN A LARGE GRADUATION AND A SMALL GRADUATION SHALL BE 0.5 MILLIMETER ± 03 MILLIMETER. SPACING BETWEEN TWO LARGE GRADUATIONS SHALL BE 1.0 MILLIMETER ± 03 MILLIMETER. THE CENTER OF THIS TEST RETICLE SHALL BE A SQUARE 2 MILLIMETER ON A SIDE WITH A USAF 1951 RESOLVING POWER TEST TARGET CONCENTRIC WITH THE CENTER OF THE TEST RETICLE. WIDTH OF ALL LINES TO BE 0127 MILLIMETER ± 0254 MILLIMETER. ALL LINES, LETTERS AND NUMBERS SHALL BE HIGH CONTRAST, BLACK ON A CLEAR GLASS SUBSTRATE 2 INCHES WIDE X 2 INCHES LONG X 060 TO 10 INCHES THICK.
2. THE USAF 1951 RESOLVING POWER TEST TARGET IN THE CENTER SQUARE IS TO BE USED FOR MEASURING CENTER RESOLUTION
3. THE FOUR EDGE USAF 1951 RESOLVING POWER TEST TARGETS ARE TO BE USED TO MEASURE PERIPHERAL RESOLUTION



25mm ASSEMBLY

FIG. 4 RESOLVING POWER TEST TARGET, USEFUL CATHODE DIAMETER, AND RESOLUTION TEST RETICLE

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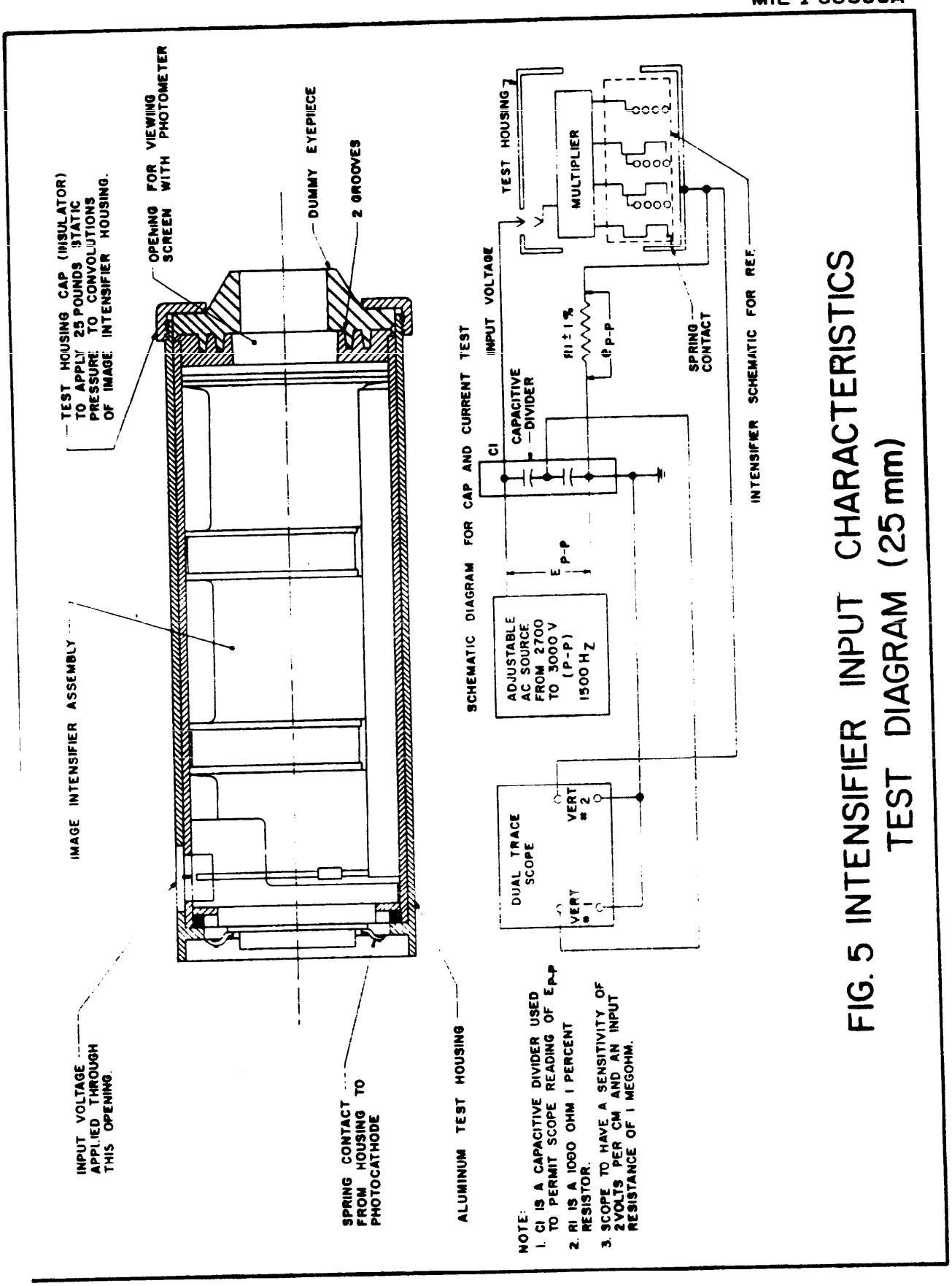


FIG. 5 INTENSIFIER INPUT CHARACTERISTICS TEST DIAGRAM (25 mm)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-I-55536A, IMAGE INTENSIFIER ASSEMBLY 25 MILLIMETER

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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